

Threat Intelligence & IOC Management

IOC Operationalization, Advanced Hunting Validation & Lifecycle Assessment

Status	Environment	Validation Window	Analyst
Completed	Defender XDR / Defender for Endpoint (SOC lab)	Last 30 days	Edwin — SOC Analyst (L1)

1. Executive Summary

This project operationalizes external threat-intelligence indicators inside Microsoft Defender for Endpoint and validates the environment against them using Advanced Hunting. Three indicators were entered — a SHA-256 associated with ValleyRAT, an IP address associated with Remcos infrastructure, and a domain associated with ClearFake infrastructure — each configured with the Audit action, High alert severity, and scope applied to All devices. The final Defender configuration shows no expiration configured for these indicators (Expires: Never), rather than a permanent enforced blocklist entry.

Targeted KQL hunts were then run against endpoint telemetry for each indicator. All three returned zero matches. This is recorded as a negative hunting result for the searched 30-day window, not as proof the indicators are universally harmless or that no historical exposure could ever have occurred (Section 7). The project demonstrates the full workflow: intelligence intake → Defender operationalization → hunting validation → documented disposition.

2. Objective

Demonstrate that threat intelligence can be converted into actionable Defender telemetry — an operational hunting and monitoring control — rather than remaining a passive list of hashes, IPs, and domains.

Capability	Demonstrated Outcome
IOC ingestion	Three threat-intelligence indicators entered into Defender
Indicator classification	Hash, IP address, and domain types represented
Action configuration	Audit action used for controlled validation
Severity & scope	High severity, applicable-endpoint scope
Lifecycle	No expiration configured (Expires: Never); lifecycle state documented for periodic review
Hunting	KQL searches executed against endpoint telemetry
Exposure assessment	No matching telemetry found in the 30-day hunting window

3. IOC Set

ID	Type	Indicator	Threat Association	Action
IOC-001	SHA-256	b3d939afd740dbde97e84a6b110c95c40873f811045686649b2d3ba1290f654f	ValleyRAT	Audit
IOC-002	IP	192.236.217.94	Remcos infrastructure	Audit

ID	Type	Indicator	Threat Association	Action
IOC-003	Domain	exclusivecloudprotection.com	ClearFake infrastructure	Audit

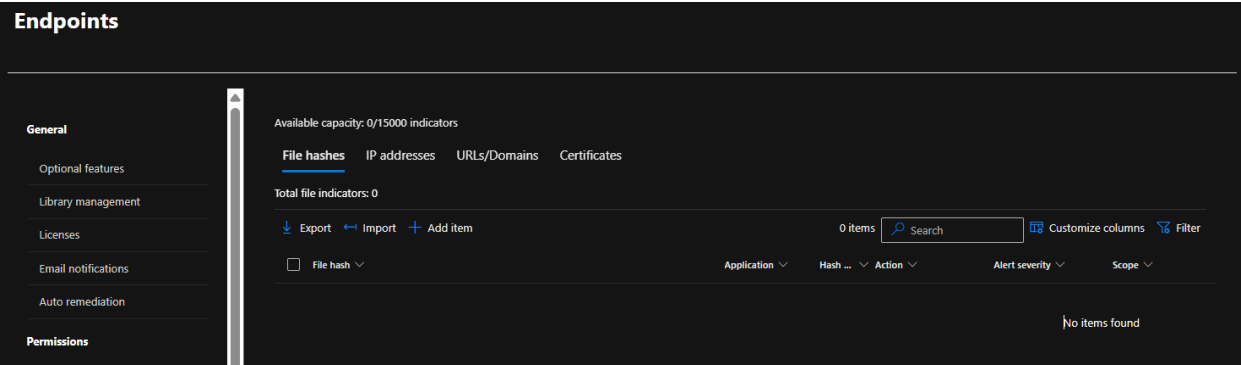
Operational Note

The Remcos indicator was operationalized as the IP address only. The associated intelligence record included a service port, but the Defender IP indicator was entered as 192.236.217.94, not an IP:port string — Defender's IP indicator type does not support port-scoped entries.

4. Indicator Operationalization

Each IOC was entered under Defender's Endpoint Indicators area using the matching indicator type. The project deliberately used the Audit action for controlled lab validation, allowing a future encounter to generate an observable alert without immediately enforcing disruptive blocking. All three indicators carry High severity and All-devices scope, with no expiration configured (Expires: Never).

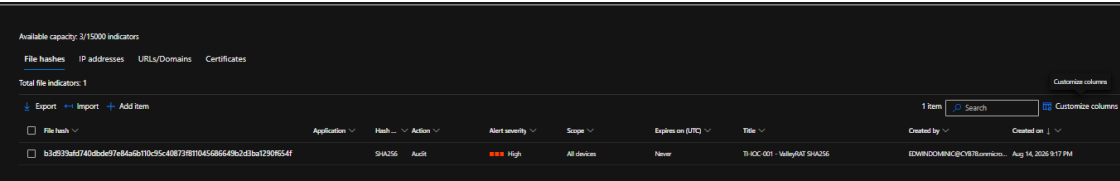
Evidence A — Baseline Indicator Inventory



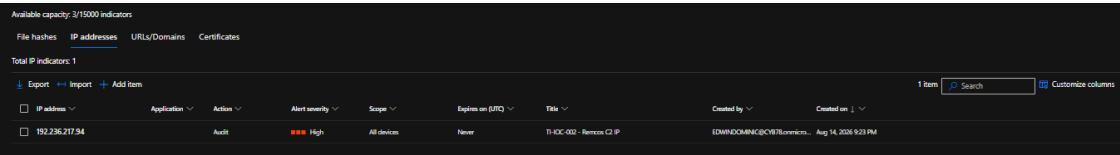
File-hash indicator view showing 0 items before operationalization — establishes the starting state of the indicator inventory.

Evidence E, F, G — Final Indicator Configuration

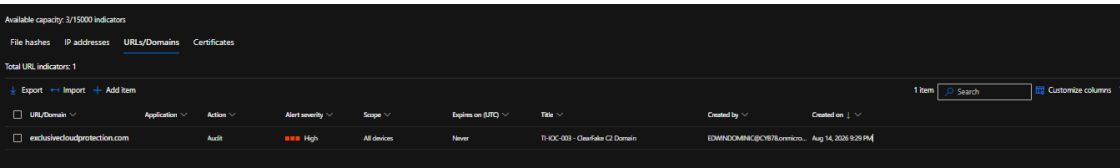
The following captures show the final, populated state of each indicator type after operationalization, confirming Action, Alert severity, Scope, and Expiration as actually deployed.



Evidence E — File hashes: TI-IOC-001 (ValleyRAT SHA-256) — Audit, High, All devices, Expires: Never.



Evidence F — IP addresses: TI-IOC-002 (Remcos C2 IP) — Audit, High, All devices, Expires: Never.



Evidence G — URLs/Domains: TI-IOC-003 (ClearFake C2 Domain) — Audit, High, All devices, Expires: Never.

5. Advanced Hunting Validation

After operationalization, three targeted KQL hunts were run against endpoint telemetry. The Advanced Hunting interface was constrained to Last 30 days, retained consistently as this project's validation window.

5.1 SHA-256 Hunt

```
DeviceFileEvents
| where SHA256 == "b3d939afd740dbde97e84a6b110c95c40873f811045686649b2d3ba1290f654f"
| project Timestamp, DeviceName, ActionType, FileName, FolderPath,
    SHA256, InitiatingProcessFileName, InitiatingProcessCommandLine
| order by Timestamp desc
```

Result: 0 items — no matching file telemetry was returned in the specified 30-day window.

The screenshot shows the 'Advanced hunting' interface in Microsoft Defender for Cloud. At the top, there's a banner for 'Secure your cloud assets with Microsoft Defender for Cloud'. Below that, a 'New query' button is visible. The left sidebar shows a schema tree with categories like Alerts & behaviors, Apps & identities, and Devices. The main area displays a KQL query: `DeviceFileEvents | where SHA256 == "b3d939afd740dbde97e84a6b110c95c40873f811045686649b2d3ba1290f654f" | project Timestamp, DeviceName, ActionType, FileName, FolderPath, SHA256, InitiatingProcessFileName, InitiatingProcessCommandLine | order by Timestamp desc`. The query is set to run against 100,000 (Default) items for the last 30 days. The results section shows '0 items' and a message: 'No results found in the specified time frame.'

Evidence B — SHA-256 hunt showing no results.

5.2 IP Hunt

```
DeviceNetworkEvents
| where RemoteIP == "192.236.217.94"
| project Timestamp, DeviceName, ActionType, RemoteIP, RemotePort,
    Protocol, InitiatingProcessFileName, InitiatingProcessCommandLine
| order by Timestamp desc
```

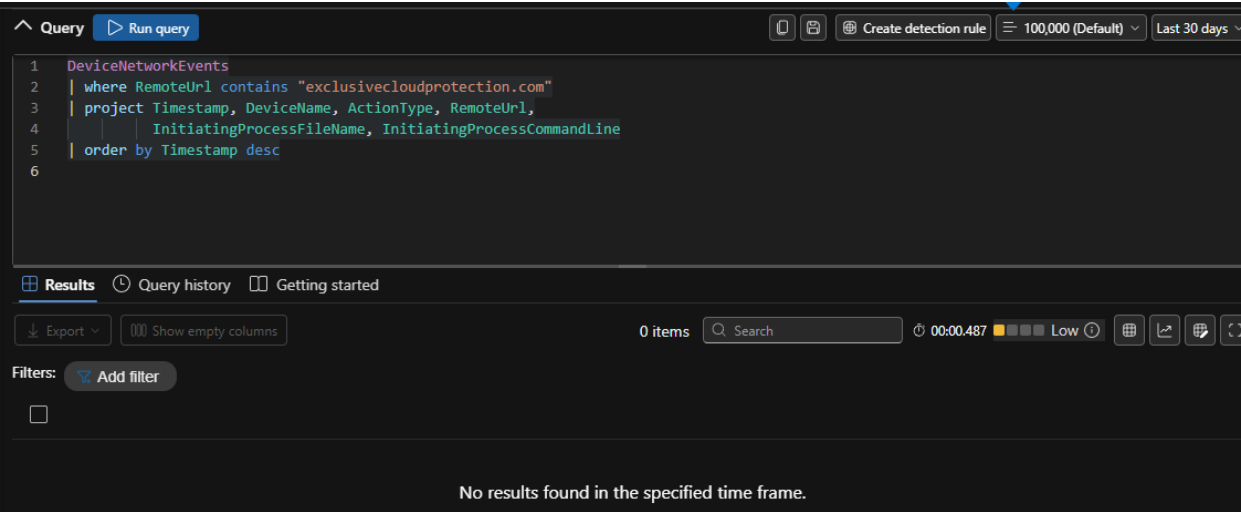
Result: 0 items — no matching network telemetry was returned in the specified 30-day window.

The screenshot shows the 'Advanced hunting' interface in Microsoft Defender for Cloud. The main area displays a KQL query: `DeviceNetworkEvents | where RemoteIP == "192.236.217.94" | project Timestamp, DeviceName, ActionType, RemoteIP, RemotePort, Protocol, InitiatingProcessFileName, InitiatingProcessCommandLine | order by Timestamp desc`. The query is set to run against 100,000 (Default) items for the last 30 days. The results section shows '0 items' and a message: 'No results found in the specified time frame.'

5.3 Domain Hunt

```
DeviceNetworkEvents
| where RemoteUrl contains "exclusivecloudprotection.com"
| project Timestamp, DeviceName, ActionType, RemoteUrl,
  InitiatingProcessFileName, InitiatingProcessCommandLine
| order by Timestamp desc
```

Result: 0 items — no matching URL/domain telemetry was returned in the specified 30-day window.



Evidence D — domain hunt showing no results.

6. Threat Context (Reference)

The table below summarizes the publicly documented threat family behind each indicator, for context only. These technique associations describe the threat family in general — they are external reference information, not evidence of technique execution observed in this environment.

Indicator	Threat Family	Typical Associated Technique(s)
IOC-001 (SHA-256)	ValleyRAT	Remote access trojan; commonly delivered via multi-stage loaders (T1105 — Ingress Tool Transfer)
IOC-002 (IP)	Remcos	Commercial RAT used as C2 infrastructure (T1071.001 — Application Layer Protocol: Web Protocols)
IOC-003 (Domain)	ClearFake	Fake browser-update delivery framework (T1204.002 — User Execution: Malicious File)

IOC Provenance

The table below records the exact-match source for each operationalized indicator value, independent of the general threat-family context above.

IOC	Exact Value	Provenance
ValleyRAT SHA-256	b3d939afd740dbde97e84a6b110c95c40873f811045686649b2d3ba1290f654f	MalwareBazaar (bazaar.abuse.ch)
Remcos C2 IP	192.236.217.94	ThreatFox (threatfox.abuse.ch) — recorded as 192.236.217.94:4551
ClearFake C2 Domain	exclusivecloudprotection.com	ThreatFox (threatfox.abuse.ch)

Threat Intelligence Sources

#	Source
1	Microsoft Defender for Endpoint — Overview of indicators (learn.microsoft.com/en-us/defender-endpoint/indicators-overview)
2	MITRE ATT&CK — Remcos, Software S0332 (attack.mitre.org/software/S0332)
3	Zscaler ThreatLabz — Technical analysis of the latest ValleyRAT variant (zscaler.com/blogs/security-research)
4	Sekoia.io — ClearFake: a newcomer to the fake updates threat landscape (blog.sekoia.io)
5	MalwareBazaar — exact ValleyRAT SHA-256 record (bazaar.abuse.ch)
6	ThreatFox — exact Remcos IP and ClearFake domain records (threatfox.abuse.ch)

7. Analyst Assessment

The investigation produced a consistent negative result across file, network-IP, and network-domain telemetry. The correct SOC interpretation is no exposure observed in the available telemetry for the searched period, not the environment is definitively clean — a negative hunt is bounded by telemetry coverage, retention, query scope, and the exact indicator representation used.

Assessment Area	Finding	Disposition
Hash exposure	0 matching DeviceFileEvents	No evidence observed
IP communication	0 matching DeviceNetworkEvents	No evidence observed
Domain communication	0 matching DeviceNetworkEvents	No evidence observed
IOC lifecycle	No expiration configured (Expires: Never)	Manually controlled / reviewable

8. Lifecycle, Limitations & Next Steps

Completed workflow: threat intelligence → IOC selection → Defender operationalization → Audit-based hunting control → Advanced Hunting → exposure assessment → documented disposition → periodic lifecycle review. This is the key competency this project adds: the analyst begins with external intelligence and turns it into an operational hunting/monitoring control, distinct from a conventional endpoint investigation that starts from an internal alert.

- Validation is limited to the Defender telemetry available within the 30-day hunting window.
- No synthetic execution of the three threat indicators was performed; this project does not claim a live positive detection.
- Next maturity step: validate one indicator in a dedicated controlled test environment, or verify an alert from a benign test event where the platform supports safe validation.
- All three indicators are currently configured with no expiration (Expires: Never). As a next step, they should be assigned a review/expiration date and periodically revalidated against current threat intelligence rather than left indefinitely active.

9. Final Project Status

Field	Value
Status	Complete
Objective met	Threat-intelligence operationalization, IOC lifecycle state assessment, and proactive hunting validation demonstrated

Field	Value
Lifecycle status	Indicators are currently configured with no expiration (Expires: Never); a future review/expiration policy was identified as the next maturity step but was not implemented during this lab
Result	Documented negative hunt across three IOC types — not a fabricated detection
Prepared by	Edwin — SOC Analyst (L1)
Scope statement	Controlled SOC lab portfolio artifact; no production incident is implied

Evidence Register

Evidence	Purpose
Baseline indicator inventory screenshot	Establishes pre-operationalization state
SHA-256 Advanced Hunting screenshot	Validates ValleyRAT hash investigation
IP Advanced Hunting screenshot	Validates Remcos C2 investigation
Domain Advanced Hunting screenshot	Validates ClearFake C2 investigation
Final Defender indicator configuration — File Hash	Proves final SHA-256 configuration (Audit, High, All devices, Never)
Final Defender indicator configuration — IP	Proves final IP configuration (Audit, High, All devices, Never)
Final Defender indicator configuration — URL/Domain	Proves final domain configuration (Audit, High, All devices, Never)